

GEOGRAPHY

INDIAN GEOGRAPHY



MINERALS & INDUSTRIES

MODULE - 8

SUGAR INDUSTRY

SUGAR INDUSTRY



LOCATION OF SUGAR INDUSTRY

9 After green revolution, the industries started shifting towards western Uttar Pradesh, Punjab, Haryana and of late industries are also moving towards peninsular India.

8 Initially, most of the industries were located in eastern parts of Uttar Pradesh and Bihar.

7 Better recovery rate of sugar is therefore dependent upon the time-period within which it has been crushed.

6 The sucrose content begins to dry during haulage after it has been harvested from the field.

5 Thus, most of the sugarcane industries are located within the cane producing region.

1 Sugar industries are located near cane cultivating areas.

2 Sugar-cane is a **weight losing raw material**.

3 The term weight losing means, the raw material once processed will be of much lesser weight than the initial product (i.e. 100 kg of raw sugarcane will be much heavier than its processed output which is the sugar).

4 Also sugarcane needs to be crushed within 24 hours of its harvesting.

SUGAR INDUSTRY

WHY IS THERE A SHIFT TOWARDS THE SOUTHERN REGION?



SUCROSE CONTENT

TEMPERATURE REQUIREMENT

- 1 Peninsular India cultivates tropical variety which has more sucrose content.
- 2 Whereas, North India has the sub-tropical variety which has less sucrose content.
- 3 Temperature required for sugarcane growth is around $22-30^{\circ}\text{C}$.
- 4 Correct range for sugarcane growth is present at Peninsular India.
- 5 Temperatures at winters in North India falls around $4-5^{\circ}\text{C}$.

SUGAR INDUSTRY

WHY IS THERE A SHIFT TOWARDS THE SOUTHERN REGION?

FAVOURABILITY OF CLIMATE



6 Sugarcane crop is a **frost averse crop** and therefore, southern parts of India are better for its cultivation.

7 **Crushing season is long** in South India which operates for around 8 months.

8 Whereas this is only 4-5 months in Northern-India i.e., maximum harvesting done at November to December in North India and as soon as it is harvested, it is crushed and sugar is produced.

9 Therefore there is only 4-5 months of operation in Northern India.

10 Whereas harvesting season is very long in South India, starting from October to April. Therefore mills operating in North India remains idle for many months and hence the mills may undergo losses.

SUGAR INDUSTRY

WHY IS THERE A SHIFT TOWARDS THE SOUTHERN REGION?



IRRIGATION FACILITIES



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Good irrigation facilities in places like Maharashtra, Cauvery Basin etc., made the farmers to shift to sugarcane.

12

After harvesting farmers need fixed market at pre-determined prices.

13

Mill owners on the other hand, need regular supply of sugarcane to operate in economies of scale.

14

Sugar industry thrives best under cooperative sector and peninsular India has better cooperative culture.

DEVELOPMENT OF COOPERATIVE SECTOR



There is a win-win situation when all this is done under co-operative sector, which is very well developed in South India. Therefore most of the mills are moving towards southern parts of India.

SUGAR INDUSTRY

WHY IS THERE A SHIFT TOWARDS THE SOUTHERN REGION?



LATEST TECHNOLOGY

- 15** Latest technology/mills is present in south India.
- 16** Therefore, there is high productivity in southern parts of India.

All this have made the area under sugarcane cultivation in Peninsular India to increase in the last 2-3 decades as sugarcane emerged as a very favorable crop in southern parts of India.

PROBLEMS OF SUGARCANE INDUSTRY

1. MONOCULTURE OF SUGARCANE



- 1 That is excessive growth of sugarcane at the cost of other crops and lack of crop rotation in some areas leads to depletion of nutrients in soil and it adversely affects the cane productivity.

- 2 Post harvest deterioration in cane quality on **account of staling and delayed crushing** contributes to low sugar recovery rate.

- 3 Sugarcane is a **water guzzling crop** and irregularity in availability of water is another major issue in the cultivation of sugarcane crop.

- 4 This has led to a **skewed water usage** which has led to a neglect of other important food crops amongst the farmers.

PROBLEMS OF SUGARCANE INDUSTRY

WHAT IS SEED REPLACEMENT RATE?

- ✓ Seed Replacement Ratio denotes how much of the total cropped area was sown with **certified seeds in comparison to farm saved seeds**.
- ✓ It refers to the percentage of seeds replaced every year by farmers.
- ✓ Due to huge demand supply gap, India suffers from a dismal seed Replacement Ratio.

2. POOR QUALITY SEEDS



- 5** Inadequate availability of quality seeds of new sugarcane varieties and poor seed replacement rate adversely affect the realisation of potential yield of sugarcane.

PROBLEMS OF SUGARCANE INDUSTRY

3. CLIMATE AND TECHNOLOGY



- 6 Further reduction in the yield is due to climatic aberrations which had led to a rise in temperature in the recent years.
- 7 The technology used by the sugarcane is obsolete and very old which makes sugar mills economically unviable.

PROBLEMS OF SUGARCANE INDUSTRY

4. COMPETITION FROM LOCAL FARM UNITS



8 Jaggery, khand seri are much in demand and it is produced by local farm units.

9 Farmers sell their produce here because they pay the price upfront and there is no delay.

10 However, the prices can be much lesser

SUGAR INDUSTRY

PROBLEMS OF SUGARCANE INDUSTRY

5. OBSOLETE TECHNOLOGY IN NORTH INDIA



6. INCREASING GOVERNMENT INTERFERENCE IN COOPERATIVES

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Example, in Maharashtra when the government releases fund to cooperative sector, it thereby influence its decision-making in an indirect manner.

PROBLEMS OF SUGARCANE INDUSTRY

7. SUSTAINABLE SUGARCANE INITIATIVE (SSI)



Sustainable Sugarcane Initiative

12 In SSI, farmers grow crop initially in a nursery for 8-10 days, then plant them in farm area with a suitable distancing method which will help to reduce water and chemical requirements by 25-30% and it increases the productivity by 20%.

13 The government should provide necessary support like training to farmers for the adoption of SSI.

PROBLEMS OF SUGARCANE INDUSTRY

8. SUGAR PRICING



Unprecedented Support to Farmers in Getting the Right Price

Cabinet approves hike in Fair and Remunerative Price (FRP) of Sugarcane



Hiked FRP of Sugarcane from ₹ 255 to ₹ 275 per quintal for Sugar Season 2018-19



Farmers to receive 77% over and above the average cost of production*



Ensure sugarcane farmers a return of more than 50% over the cost

*cost of production estimated to be ₹ 155 per quintal



- 1 Another crucial issue with regard to sugar crisis is the pricing problem of sugarcane.
- 2 Government, in a way, is responsible for this sugar crisis. The Central Government announces Fair and Remunerative price (FRP). The states too, from their side, give more concession in the form of state advised price (SAP).
- 3 The SAP will be generally higher. This is done to fulfill the vote bank issues as sugarcane farmers form a large vote bank. For example, in one financial year, in Uttar Pradesh FRP was Rs. 240 and SAP was Rs. 305.

SUGAR INDUSTRY

PROBLEMS OF SUGARCANE INDUSTRY

8. SUGAR PRICING

LET US LOOK AT THE EXAMPLE OF THE
YEAR 2017-18



1 The market prices of sugar were hovering around Rs. 27 per kg.

2 The cost of production for the sugar mill owners is set around Rs. 34 per kg.

3 So, mills by paying FRP are incurring a loss of about Rs. 7 per kg on the sugar.

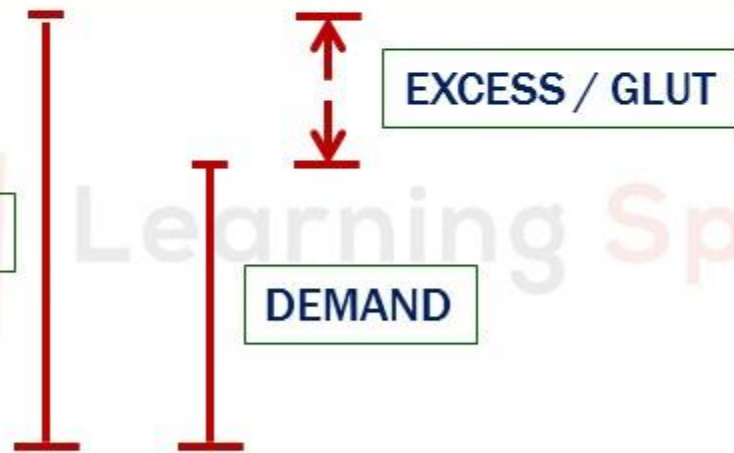
4 This has led to an increase of arrears to be paid to the farmer to the tune of Rs. 22,000 Cr.

5 Also, because of the high FRP, more and more farmers have indulged in the cultivation of sugarcane.

LET US LOOK AT THE EXAMPLE
OF THE YEAR 2017-18

8. SUGAR PRICING

- 6 Because of which
- Supply ... 32.5 million tonnes
 - Demand ... 26 million tonnes



- 7 In short, the supply has exceeded the demand because of which there is an excess of sugar in the market which has led to a glut like situation.



In 2018-19, the production is likely to go down to 30 million tonnes. Still, it is much higher than the demand. Moreover, the mills are holding huge carry-over stocks.

SUGAR INDUSTRY

PROBLEMS OF SUGARCANE INDUSTRY

8. SUGAR PRICING

LET US LOOK AT THE EXAMPLE OF THE YEAR 2017-18

- 8** Also to avoid cheap sugar from international market creeping in to India, the import duty is imposed.



- 9** In spite of all this, farmers are not profited because the mill owners are not paying the farmers upfront. There is a huge dues that has been piled up.

- 10** The recent protests by farmers of Maharashtra proves the fact that how the sugar industry is always on the boil with cash strapped mills and angry farmers without their payments.

Because of these reasons, there is interference at government in every stage – at the stage of price fixing, byproduct sale and fixation of some other criteria.

SUGAR INDUSTRY

THE MAJOR REGULATORY ISSUES AND RANGARAJAN COMMITTEE RECOMMENDATIONS

CANE RESERVATION AREA

MINIMUM DISTANCE CRITERIA

CANE PRICING POLICY

LEVY SUGAR MECHANISM

IMPORT-EXPORT POLICY

PRICING CONTROL OVER
SUGAR BY-PRODUCTS

JUTE PACKAGING

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